

Image & Video Generation

Stable Diffusion + LoRA + 20 sample photos = infinite fashion-catalogue content. Indian e-commerce saved ₹200 crore of photography costs in 2025. Today we build that capability.

How many product photos did your last employer shoot per month?

L5

WHAT'S INSIDE

- 01 The one-sentence intuition
- 02 Forward + reverse
- 03 15-line Stable Diffusion
- 04 LoRA for diffusion
- 05 2026 landscape

WHY THIS LESSON MATTERS

Nykaa generates 10 lakh product images a month. Zero photoshoots.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

- 1 Forward + reverse processes.
- 2 Use Stable Diffusion.
- 3 LoRA fine-tune on custom images.
- 4 SD / FLUX / DALL-E / Sora.
- 5 Classifier-free guidance.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

The one-sentence intuition

Add noise, then reverse

02

Forward + reverse

Fixed vs learned

03

15-line Stable Diffusion

diffusers pipeline

04

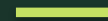
LoRA for diffusion

Custom subject from 10 images

01

PART 1 OF 4

The one-sentence intuition



Add noise, then reverse

The one-sentence intuition

Add noise to an image over 1000 steps → pure static.

Train a network to reverse each step.

At inference: noise → reverse chain → image.

Hard generation problem → sequence of easy denoising problems.

TRICK

Easy denoising

Beats direct generation.

DEFINITION

The one-sentence intuition

Add noise, then reverse

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

Forward + reverse

FORWARD $q(x_t | x_{t-1})$ — add Gaussian noise. Fixed.

REVERSE $p(x_{t-1} | x_t)$ — U-Net learns to denoise.

Train: sample t ; noise $x_0 \rightarrow x_t$; predict the noise.

Sample: 20-50 steps (DDIM, flow-matching schedulers).

KEY

Reverse is learned

Forward is math.

KEY POINTS

- FORWARD $q(x_t | x_{t-1})$ — add Gaussian noise. Fixed
- REVERSE $p(x_{t-1} | x_t)$ — U-Net learns to denoise
- Train: sample t ; noise $x_0 \rightarrow x_t$; predict the noise
- Sample: 20-50 steps (DDIM, flow-matching schedulers)

15-line Stable Diffusion

```
StableDiffusionPipeline.from_pretrained(...)  
pipe("a Banarasi saree, soft light, fashion photo")  
num_inference_steps=30, guidance_scale=7.5
```

Save .png; share.

USE IT

15 lines

Image gen in one lecture.

DEFINITION

15-line Stable Diffusion

diffusers pipeline

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

LoRA for diffusion

DreamBooth + LoRA training.

10-20 subject images + captions.

500-1000 training steps on Colab T4.

Output: 50-200 MB LoRA file.

Plug into base pipeline; generate infinite variations.

CUSTOMISE

10 images

5 minutes of training.

KEY POINTS

- DreamBooth + LoRA training
- 10-20 subject images + captions
- 500-1000 training steps on Colab T4
- Output: 50-200 MB LoRA file

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Your first generated image

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1

2

3

4

Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Where does your org spend most on photography?

Q2

What would 10,000 on-demand images unlock?

Q3

What ethical line would you NOT cross with generation?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

Forward process:



Learned



Adds noise; fixed



Denoises



Random

Forward process:

CORRECT ANSWER



Adds noise; fixed

WHY

Forward is mathematical noising; reverse is learned

Typical 2026 sampling steps:

A B

B D

C Answer: B. DDIM/Euler/flow-matching cut 1000 to 20-50.

D (no option)

Typical 2026 sampling steps:



CORRECT ANSWER

D

WHY

The correct option matches the lesson's core principle.

LoRA for diffusion size:

A

1 KB

B

1 MB

C

50-200 MB

D

10 GB

LoRA for diffusion size:



CORRECT ANSWER

50-200 MB

WHY

Small enough to ship; large enough to capture identity

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Add noise to an image over 1000 steps → pure static.

Train a network to reverse each step.

At inference: noise → reverse chain → image.

Hard generation problem → sequence of easy denoising problems.

TRICK

Easy denoising

Beats direct generation.

THE TAKEAWAY

"Everything else is tuning."

WHAT TO NOTICE

1

The one-sentence intuition

2

Forward + reverse

3

15-line Stable Diffusion

4

LoRA for diffusion

Fine-tune on Banarasi sarees

15 sample photos.

DELIVERABLES

- Collect + caption.
- LoRA train 500 steps.
- Generate 5 lifestyle images.
- Post.

WHAT 'GOOD' LOOKS LIKE

Deliverable: 5 images on forum.

ONE THING TO REMEMBER

“

"Everything else is tuning."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 6 — Edge AI.